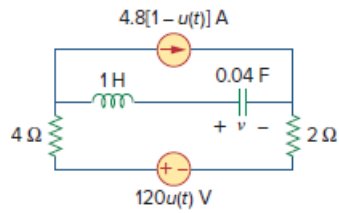


Problem

For the circuit in Fig. 16.51, find $v(t)$ for $t > 0$.

Figure 16.51:



Step-by-step solution

Step 1 of 6

Recall the definition of the unit step function.

$$u(t) = \begin{cases} 1; & t \geq 0 \\ 0; & t < 0 \end{cases}$$

The circuit in Figure 1 is in steady state at $t = 0^-$.

Replace the inductor with short circuit and the capacitor with open circuit and draw the equivalent circuit as shown in Figure 1.

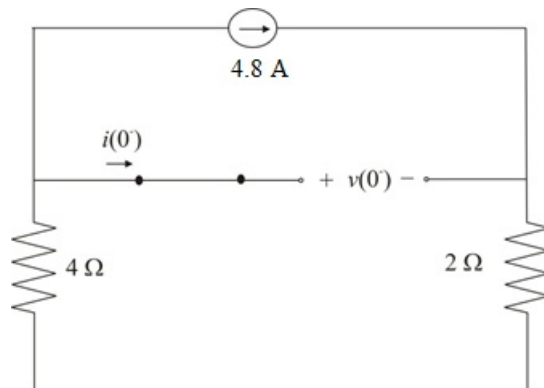


Figure 1

Step 2 of 6

The inductor current at $t = 0^-$ is,

$$i_L(0^-) = 0 \text{ A}$$

The voltage across the capacitor at $t = 0^-$ is,

$$\begin{aligned} v(0^-) &= -2 \times 4.8 - 4 \times 4.8 \\ &= -28.8 \text{ V} \end{aligned}$$

Step 3 of 6

Convert the circuit into Laplace domain as shown in Figure 2.

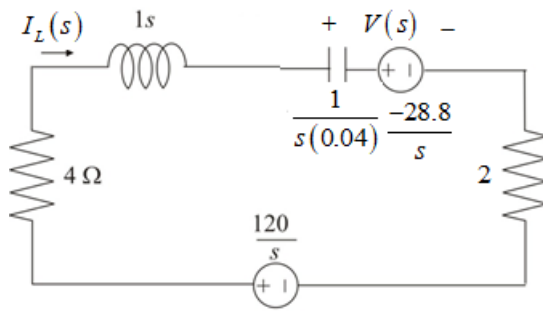


Figure 2

Step 4 of 6

Apply Kirchhoff's voltage law in the circuit shown in Figure 3.

$$4I_L(s) + sI_L(s) + \left(\frac{1}{s(0.04)}\right)I_L(s) - \frac{28.8}{s} + 2I_L(s) = \frac{120}{s}$$

$$4I_L(s) + sI_L(s) + \left(\frac{1}{s(0.04)}\right)I_L(s) + 2I_L(s) = \frac{148.8}{s}$$

$$I_L(s) \left(6 + s + \frac{25}{s}\right) = \frac{148.8}{s}$$

$$I_L(s) \left(\frac{s^2 + 6s + 25}{s}\right) = \frac{148.8}{s}$$

Simplify further.

$$I_L(s) = \frac{148.8}{s^2 + 6s + 25}$$

The voltage across the capacitor is,

$$\begin{aligned} V(s) &= \left(\frac{1}{0.04s}\right)I_L(s) - \frac{28.8}{s} \\ &= \frac{25}{s} \left(\frac{148.8}{s^2 + 6s + 25}\right) - \frac{28.8}{s} \\ &= \frac{3720}{s(s^2 + 6s + 25)} - \frac{28.8}{s} \end{aligned}$$

Step 5 of 6

Determine the partial fraction expression for $\frac{3720}{s(s^2 + 6s + 25)}$.

$$\frac{3720}{s(s^2 + 6s + 25)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 6s + 25}$$

$$3720 = A(s^2 + 6s + 25) + s(Bs + C)$$

Compare the coefficients of s^0 on both sides.

$$25A = 3720$$

$$A = 148.8$$

Step 6 of 6

Compare the coefficients of s^2 .

$$0 = A + B$$

$$B = -A$$

$$B = -148.8$$

Compare the coefficients of s^1 .

$$6A + C = 0$$

$$6(148.8) + C = 0$$

$$C = -892.8$$

Recall the expression of $V(s)$.

$$\begin{aligned} V(s) &= \frac{3720}{s(s^2 + 6s + 25)} - \frac{28.8}{s} \\ &= \frac{148.8}{s} + \frac{-148.8s - 892.8}{s^2 + 6s + 25} - \frac{28.8}{s} \\ &= \frac{120}{s} - 148.8 \frac{s}{(s+3)^2 + 4^2} + 223.2 \frac{4}{(s+3)^2 + 4^2} \end{aligned}$$

Apply inverse Laplace transform.

$$v(t) = 120u(t) - 148.8e^{-3t} \cos(4t)u(t) + 223.2e^{-3t} \sin(4t)u(t)$$

Thus, the expression of $v(t)$ is $\boxed{120u(t) - 148.8e^{-3t} \cos(4t)u(t) + 223.2e^{-3t} \sin(4t)u(t)}.$